H-PSEUDO re-read against a negation-matched null

Lead: LEAD-EXO-dl-character-sum-pseudorandom (H-PSEUDO). The original run is EXP-ECDLP-1ed445 / RUN-ECDLP-1ed445-001 (EV-ECDLP-f83a17). Protocol protocol.json, HPSEUDO-SYMNULL-20261104, was frozen before data (sha256 cf05250833d8..., commit f9931a3e).

Verdict (frozen rule): EXPLAINED, with every control valid. H-PSEUDO's recorded constant offset R ≈ 1.28–1.34 is fully accounted for by a null mismatch. The original run keys points by (x, y), so the small-x factor base is closed under negation and its character sums are real. The null was an asymmetric uniform subset. Against a negation-closed null, R is **consistent with 1**. The upper 95% bound is 1.09 at 2^20 and 1.02 at 2^22 (out of sample), and the pooled 56 curves give 1.005 ± 0.008. The statistic is the B = 512 maximum character sum of the small-x set only, and the frozen rule is a non-rejection rule, so a residual of up to about +9% at 2^20 is not excluded. (Wording corrected after independent validation, VALIDATION-round3.md.)

Design

- **EC arm.** The recorded per-curve C_EC from stage2.per_curve, with B = 512. This study does not re-measure them; it replays a subset instead (below).
- **Decisive (out-of-sample) curves.** At bits 20 and 22, the 12 recorded curves that did not feed R_mean (curve_index 8–19). No matched-null ratio had been computed for them before this protocol. **Calibration curves:** the 8 R_mean curves at each rung. These were already re-read in-sample by the round-2 validator, and the protocol checker rejected them as decisive for that reason.
- **Nulls.** Each curve's own N gets 20 negation-closed random 512-sets. The original law, uniform 512-subsets with 40 draws at each rung's representative N, is used as a replication check against the recorded null means.
- **Controls.**
  - Positive: a planted negation-closed arithmetic progression in log space.
  - Replication: the asymmetric null mean against the recorded C_null_mean.
  - Replay: C_EC recomputed with the original code (run_character_sum_ladder.py) for 18 curves, including the smallest-N curve at bits 24 and 26.
  - A/A: even- against odd-indexed draws.
- **Rule.** EXPLAINED iff at both decisive rungs the bootstrap CI of mean R_sym contains 1 and |mean R_sym − 1| ≤ 0.05.

Results

rung	recorded R (asymmetric null)	R_sym, calibration (8 curves) [95% CI]	R_sym, out of sample [95% CI]	mean u_sym: calibration / out of sample
20	1.334	1.012 [0.963, 1.061]	1.040 [0.999, 1.092] (n=12)	0.55 / 0.66
22	1.280	0.987 [0.954, 1.032]	0.993 [0.972, 1.017] (n=12)	0.36 / 0.47
24	1.275	0.974 [0.948, 1.000]	–	0.35 / –
26	1.342	1.009 [0.990, 1.036]	–	0.57 / –

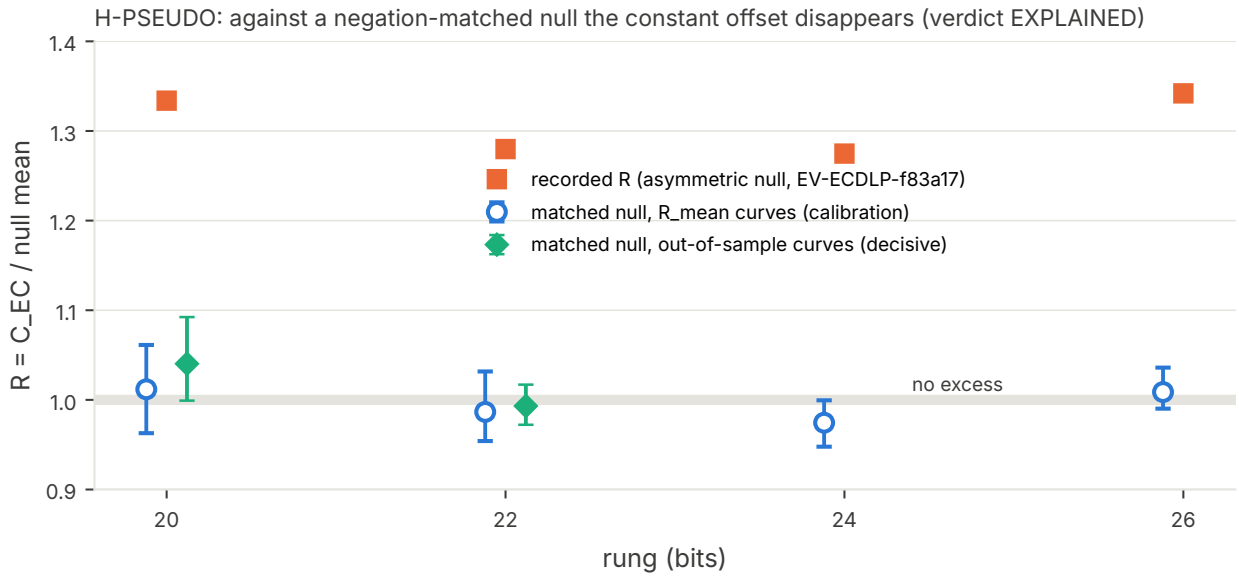

Figure 1: R against matched null

- **Validity, all passed.**
  - Replication: the asymmetric null reproduces the recorded C_null_mean to within 0.94% at every rung.
  - Positive control: the planted AP is 3.5–4.0× the rung's largest symmetric draw.
  - Replay: all 18 replayed curves reproduce the recorded C_EC within 1e-5, with negation-closed factor bases.
  - A/A: within 2%.
- **Decisive result.** The out-of-sample rungs give R_sym = 1.040 (2^20) and 0.993 (2^22), with CIs containing 1. The 2^20 interval only just reaches 1 (lower end 0.999), so a residual of a few percent at that rung is not excluded. 2^22 is centred on 1.
- **Calibration rungs.** These span 0.974–1.012, matching the round-2 validator's in-sample recomputation (0.96–1.03).

Consequence

The H-PSEUDO row's best_measured (R ≈ 1.28–1.34) and its framing "the curve behaves like a random set up to a constant" are corrected. Against a matched null, the recorded excess disappears. The B = 512 character-sum maximum of the small-x set is consistent with a random negation-closed set at 2^{20-2}26 (upper bounds above). The 2^24 calibration interval [0.948, 0.9995] lies slightly below 1. The disposition exhausted_at_scale is unchanged. The revisit trigger should name the matched null. Any other comparison of a negation-closed factor base against an asymmetric null needs the same correction.

Deviations

- **Out-of-memory hang (infrastructure).** The memory cgroup killed the three pool workers, at about 5 GB each, on the N ≈ 33.6M prime-length FFTs (dmesg). The pool then hung with 36 null tasks lost. complete_run.py recomputed exactly those tasks with the frozen job() and seeds, in one process, then ran the frozen control and replay stages, whose code is copied verbatim. No design parameter changed. This is the protocol's infrastructure-only retry.
- **No S3 checkpoint.** There is no upload token. With the owner's approval, the pushed branch is the resume record.
- **Compression.** Rows are stored as .xz. The uncompressed hash is in data.sha256.

Reproduce

xz -dc rows.jsonl.xz > rows.jsonl # or rerun (needs ~6 GB per worker at the top rung; use --workers 1 or 2):
python3 hpseudo_symnull.py --protocol-sha cf05250833d83b48fa05f8df8e37a0f3f2ff71e0da4d14b15c790d428987b135 --out rows.jsonl --workers 2
python3 analyze_symnull.py rows.jsonl verdict.json
python3 figures.py; ./build_pdf.sh